

TECHNICAL SKILLS

Languages: C/C++, Python, C#/.NET, MATLAB, HTML/CSS, R, VHDL, Assembly

AI & Frameworks: YOLO, OpenCV, OCCT, PyTorch, Hugging Face

Development Tools : Git, CMake, Docker, Linux, Visual Studio, LTSpice, Fusion 360

Hardware: Raspberry Pi, Arduino, Microcontrollers, FPGA, I2C, UART/Serial, PCB Design, Oscilloscope, Soldering

EDUCATION

University of British Columbia, BaSc Integrated Engineering 09/2023 – 04/2028

- Major: Computer Engineering, Minor: Electrical Engineering, GPA: 3.5
- Co-op: Available for 4/8/16 months beginning of Fall 2026

TECHNICAL WORK EXPERIENCE

UBC Concrete Canoe, Software Team 09/2025 – present

- Architecting a simulation software for different concrete canoe designs in C++ using OCCT
- Implemented wire-frame logic and mesh rendering

UBC Rapid, Filament Recycler - Electrical Member 09/2024 – 01/2025

- Drafted a wiring diagram connecting the extruder, housing, and temperature control
- Assisted with maintenance of existing 3D printers to provide accessible service for other students

WORK EXPERIENCE

Lee's Donuts, Decorator/Cashier 04/2026 – present

- Optimized workflow and decorated up to 2000 donuts with high attention to detail.
- Managed high-volume transactions and delivered rapid, exceptional service during peak hours.

Apricity Cashmere, Sales Associate 04/2024 – present

- Launched an e-commerce and social media strategy, expanding North American reach and boosting sales by 20%.
- Provided personalized, consultative service to 40+ daily customers.

Steveston Bakery, Cashier/Server 03/2023 – 08/2023

- Processed transactions and fulfilled orders for 100+ daily customers in a fast-paced environment.
- Coordinated with kitchen staff to streamline operations and ensure accurate, timely service.

TECHNICAL PROJECTS

Wearable Obstacle Alert Device 09/2025 – 04/2026

- Ongoing project: creating a device to assist visually impaired people detect dangerous obstacles
- Trained a custom YOLO computer vision model using Roboflow and Google Colab
- Developed a program in Python with a camera, ToF sensor and a CV model for detection and feedback in real time
- Implemented program into Raspberry Pi and went through testing and debugging for maximum optimization

Autonomous Storage and Retrieval System 01/2025 – 04/2025

- Worked closely with 5 students to build a system and took the lead of software integration
- Coded a web-interface with embedded C in CSS and HTML
- Troubleshooted and incorporated 4 motors into 3 axis of movements via a web server hosted on ESP-32
- Incorporated controls for autonomous functions, such as placement of units and the storage and retrieval actions

Autonomous Line-Following Robot 09/2024 – 01/2025

- Designed a bracket to insert the IR sensors in AutoDesk Fusion 360 and 3D Printed the structure
- Designed the PCB structure that connected the power, sensor signal, motors, and Arduino UNO and soldered the pieces
- Troubleshooted and reviewed the algorithm and code (in C) for following a black line using 5 IR sensors
- Successfully completed 3 different levels of track autonomously.

VOLUNTEER/CLUBS

UBC Agora Cafe, Community Dinner Volunteer 09/2025 – present

UBC Photo Society, Magazine Designer 09/2024 – 04/2025

SOS Children's Village Thrift Store, General Labour/Cashier 08/2023 – 02/2024